SCHOOL OF COMPUTING AND DIGITAL TECHNOLOGY

M.Sc. Computer Science

CMP7200

Individual Master's Project (Project Proposal)

Title: Exploring Automatic Text Summarization Using a Modular Pipeline Approach with LangChain

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Submission Date: October 21st, 2024

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Abstract

Automatic text summarization is increasingly important for managing and condensing large volumes of information, such as news articles, research papers, or reports but still, current models often struggle to balance accuracy, coherence, and fluency in the summaries they produce. This project explores an innovative modular pipeline approach for text summarization using LangChain, a framework that enables seamless integration of multiple models and pre/post-processing steps. With more flexibility and customisation possible, this approach uses many models from hugging face to improve the overall quality of summaries.

The research focuses on experimenting with various combinations of summarization models, refining the output at different stages, and evaluating their effectiveness using both automated metrics (ROUGE, BLEU, BERTScore) and human assessments. The modular pipeline not only combines the strengths of different models but also targets to find limitations like redundancy, fluency, and factual accuracy. This approach opens possibilities for domain-specific applications, where text Summarization can be tailored for specialized tasks.

This project seeks to explore whether a multi-step, model-agnostic pipeline could lead to improved performance and offer a more adaptable solution for real-world summarization needs.

Keywords: Automatic text summarization (ATS), modular pipeline, LangChain, hugging face models, natural language processing (NLP), summarization quality, pre-processing, post-processing, ROUGE, BLEU, BERTScore, multi-model integration, real-world applications, human evaluation, text coherence, fluency, factual accuracy.

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1. Introduction

1.1. Background

In the age of digital information, the rapid expansion of textual data plays an enormous challenge for users to efficiently access and comprehend relevant information. With the internet and various textual sources generating massive amounts of data daily, finding concise and accurate summaries is becoming increasingly essential. Automatic Text Summarization (ATS) is a solution that enlightens this challenge by filtering essential information from large datasets into brief, manageable summaries (Patel & Tabrizi, 2022; Widyassari et al., 2020). However, as the field of text Summarization evolves, different approaches—such as extractive and abstractive methods—have emerged, each with its advantages and limitations (Andhale & Bewoor, 2022).

ATS is significant in domains like news aggregation, scientific research, and business intelligence, where summarizing vast amounts of data quickly and accurately is vital. Traditional extractive methods, which select important sentences or phrases directly from the source, have been widely used due to their simplicity and speed. However, these approaches often lead to summaries that lack coherence and fluidity, as they rely on exact sentences from the original text (Andhale & Bewoor, 2022; Widyassari et al., 2020). Conversely, abstractive methods aim to generate new sentences based on the original content, resembling human-like summaries that are more fluid and natural but come with higher computational complexity (Zhang et al., 2022; Shukre et al., 2023).

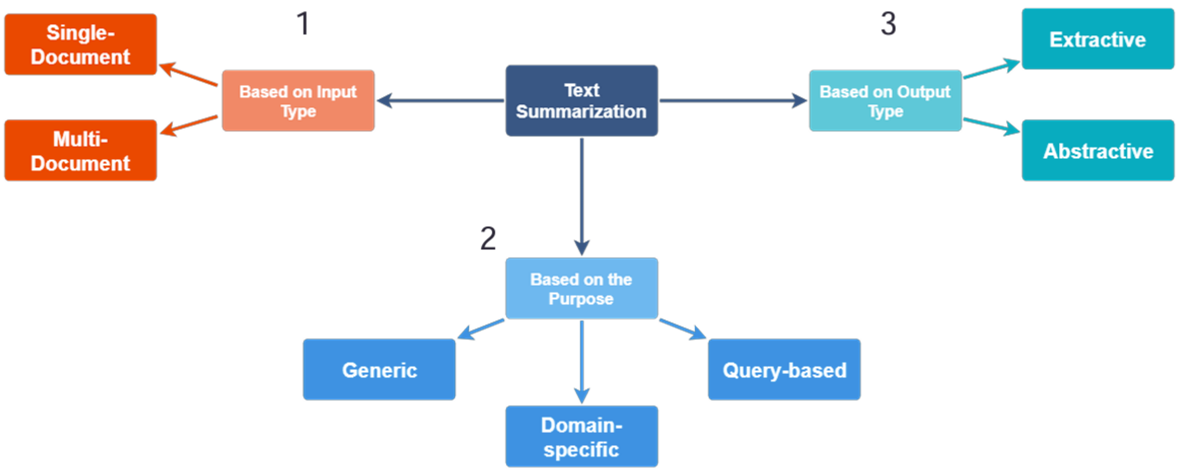

Figure 1 Types of Text Summarization (Vishwa Patel et al, 2022)

Figure 1 illustrates that text summarization methods are classified based on input, output and purpose. Based on the input the summarization process handles one document or synthesizes information from multiple documents (Chauhan K, 2018) whereas based on purpose, it deals with domain-specific text and queries (Vishwa Patel et al, 2022). This research is going to deal with based on output which is extractive and abstractive. A key limitation of both approaches lies in balancing speed and quality. Extractive Summarization is faster but often produces mechanical summaries, while abstractive Summarization, though more accurate in terms of meaning, is computationally intensive and prone to generating factual errors (Widyassari et al., 2020; Zhang et al., 2022). The quality of abstractive Summarization has improved due to available developments in neural networks, especially in deep learning-based models like

Transformer architectures (e.g., BART and Pegasus), but the models still struggle with generalizing across different datasets and domains (Zhang et al., 2022). Thus, there is a growing need for flexible, modular approaches that combine the strengths of both methods while overcoming their weaknesses.

1.2. Rationale

Automatic Text Summarization (ATS) has shown tremendous growth but still faces obstacles where one of the key limitations of current models is the trade-off between speed and quality. Extractive summarization, though fast and straightforward, often generates summaries that are disjointed and lack coherence (Widyassari et al., 2020). On the other hand, abstractive summarization models offer more fluent and human-like summaries but come with issues like hallucination, where generated content deviates from the source (Maynez et al., 2020). Recent transformer models such as BART and Pegasus have improved performance but continue to struggle with maintaining factual accuracy across varied domains (Zhang et al., 2022). This project aims to address these gaps by exploring a modular pipeline approach, combining extractive and abstractive methods to leverage their strengths while mitigating their weaknesses. Recent research highlights those modular systems, which incorporate multiple specialized models, can improve factual consistency and adaptability across domains (Li et al., 2023). Using LangChain to implement this approach offers flexibility for experimenting with different model combinations and workflows. Additionally, pre-trained models from Hugging Face allow rapid prototyping, reducing development time while improving performance (Zhang et al., 2024). This study contributes to bridging the gap by proposing a flexible, hybrid solution to enhance summary quality and relevance across different datasets and domains.

Figure 2 below illustrates a text summarization pipeline using Streamlit and LangChain. A user inputs text through a web interface provided by Streamlit. This input is passed into LangChain's Character Text Splitter, which divides the text into manageable chunks before converting them into Document() objects. These documents are then processed by the load_summarize_chain(), which utilizes a Large Language Model (LLM) to generate a summarized version of the input text. The output is displayed back to the user through the Streamlit interface, creating a real-time summarization experience.

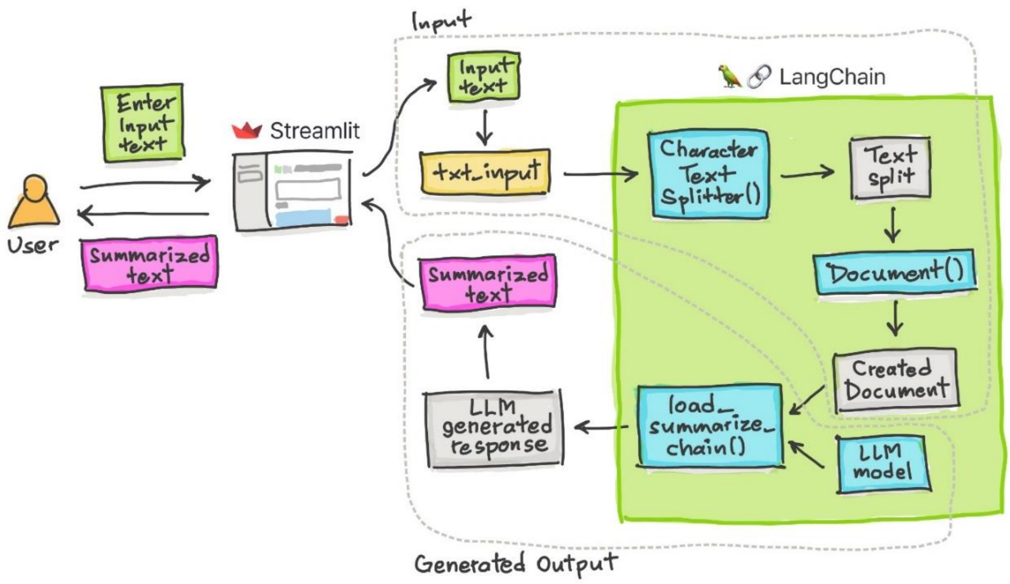

Figure 2 Summarization Pipeline (Jyoti Dabass, 2024)

Figure 3 below demonstrates a multi-step pipeline for summarization using a Large Language Model (LLM). The user input passes sequentially through multiple processing chains, each refining or altering the input. The output of each chain feeds into the next until the final chain produces the result. Throughout this process, the LLM interacts at different stages, contributing to each step. This pipeline setup shows how a complex, multi-step approach can be used to handle tasks such as summarization by progressively refining the output.

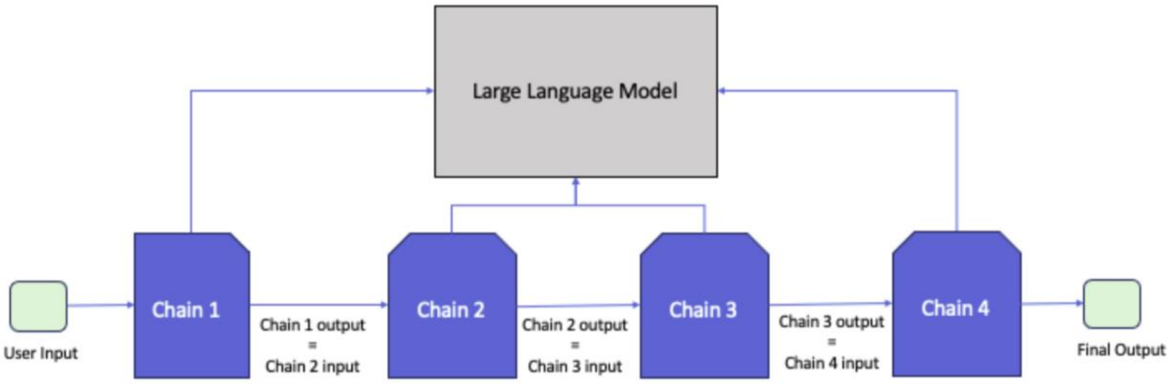

Figure 3 LangChain Functioning (Bhaskar Dhavan, 2023)

Figure 4 illustrates a workflow for integrating a Large Language Model (LLM) with various data sources using the LangChain framework. The process begins with raw data sources, where external data is fetched for processing. The data pipeline handles pre-processing tasks, such as generating word embeddings. These embeddings are stored in a vector store, enabling efficient retrieval when needed. The LangChain framework plays a pivotal role, interfacing with various LLM providers. It creates prompt templates, generates prompts for the model, and manages the input/output processes. Prompt engineering is used to craft appropriate inputs for the LLM, ensuring the model can generate meaningful outputs. Once the LLM processes the input, the resulting output is sent back to the user through a user interface. This pipeline demonstrates how LangChain connects data, preprocessing, and LLMs to enable advanced language model applications.

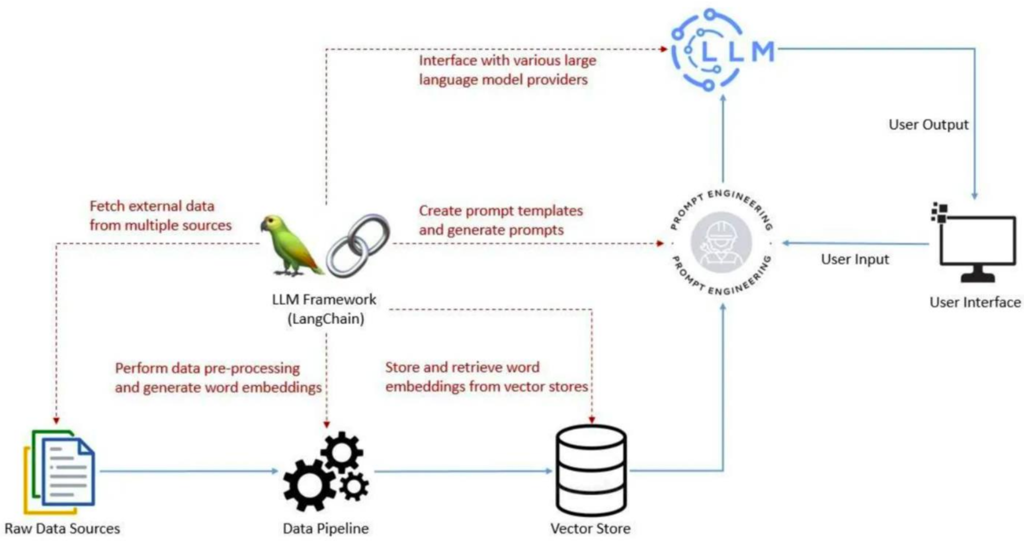

Figure 4 Power of LangChain to LLMS (Nil Makvana, 2024)

1.3. Aim

To explore the effectiveness of a modular pipeline approach for automatic text Summarization by leveraging multiple models and tools through LangChain, with the goal of improving the quality and efficiency of the Summarization process.

1.4. Objectives

The following goals will assist in achieving the study's primary goal:

1. Investigate and learn existing methods for automatic text Summarization, focusing on the strengths and weaknesses of both extractive and abstractive approaches.

2. Develop a flexible and modular pipeline using LangChain to combine multiple models and techniques for efficient text Summarization.

3. Test and analyze various Hugging Face model combinations to explore their effectiveness in improving Summarization performance.

4. Utilise human assessment in addition to recognised metrics such as ROUGE, BLEU, and BERTScore to evaluate and improve the quality of the generated summaries.

1.5. Research Question

The following research questions will direct this study:

1. How does the performance of a modular pipeline approach for automatic text Summarization compare to traditional single-model approaches?

2. How do different pre-processing and post-processing steps within a modular pipeline affect the overall Summarization quality?

3. Can a modular pipeline approach be effectively adapted for specific domains (e.g., legal, medical, or social media) to improve Summarization outcomes?

2. Literature Review

Automatic text summarization is one of the famous tools widely used to concisely summarize large volumes of information while retaining the key points, categorized primarily into extractive and abstractive methods. The goal of these techniques is to distil vast amounts of information while keeping the essential elements. Abstractive summarization creates new sentences where the idea is taken from the original text, whereas extractive summarization chooses important sentences or phrases straight from the text (Ladhak et al., 2022). Development in NLP and deep learning resulted in shifting to abstractive summarization.

2.1. Advancements in Text Summarization

Transformer models have significantly exceeded traditional rule-based or statistical methods in both speed and quality (Torbarina L et al., 2024). The transformer-based models like BERT and GPT have shown an upgrade whereas models like BART (Lewis et al., 2020) and T5 (Raffel et al., 2020) have been shown to achieve innovatory results in both extractive and abstractive summarization tasks.

However, these models are not without challenges. Hallucination in summarization—where the model generates content that is not in the original text—is a notable issue with abstractive models (Maynez et al., 2020). Despite advancements, many current models struggle with maintaining factual consistency and producing summaries that are fully aligned with the source material (Ji Z et al., 2022). This has led to growing interest in modular or hybrid approaches to summarization, where different models or techniques are combined to mitigate these limitations.

2.2. Modular Pipeline Approaches in NLP

A modular pipeline approach is one in which multiple models or components are chained together, each responsible for a specific task within the overall process. This concept is gaining attention in NLP due to the increasing availability of specialized models that perform well on specific subtasks. For example, LangChain enables the integration of multiple models and techniques within a pipeline, facilitating both extractive and abstractive summarization workflows. According to recent research, combining multiple models—such as using an extractive summarizer to select important sentences followed by an abstractive model to rephrase them—can reduce hallucination and improve factual consistency (Li et al., 2023). LangChain's flexibility in chaining different models and allowing custom steps offers opportunities to experiment with various model architectures. For instance, an extractive summarizer can serve as a filter to identify key parts of a text, after which a transformer-based abstractive model generates a more refined summary. Research has indicated that the integration of extractive and abstractive strategies can improve the factual correctness and fluency of summaries (Gong et al., 2022). Furthermore, pipelines are more adaptive to context-specific requirements since they may be customised to domains (Huang et al., 2023).

2.3. Hugging Face and Model Combinations

Hugging Face provides a vast collection of models, including those specialized for summarization tasks. This enables researchers to experiment with different model combinations without needing extensive computational resources for training from scratch. Research by Zhang et al. (2024) highlights that using pre-trained Hugging Face models in a modular pipeline, particularly with tools like LangChain, can drastically reduce development time and improve performance by allowing rapid prototyping of different model combinations. Recent studies have explored the benefits of combining Hugging Face models with task-specific pipelines. Zhang et al. (2024) show that integrating multiple models for tasks like sentence ranking (extractive) and paraphrase generation (abstractive) improves the summarization pipeline's output quality. This modular approach allows for experimentation and fine-tuning, enabling researchers to adapt models to specific datasets and domains, thereby creating a more versatile and robust summarization system.

2.4. Evaluation Metrics for Text Summarization

Generating a good summary is not easy in text summarization. The base or regular metrics like ROUGE (Lin, 2004) and BLEU (Papineni et al., 2002) have been widely used to measure the n-gram overlap between the generated summary and a reference summary. These metrics provide a quick and quantitative way to evaluate summarization systems, but when it comes to semantic similarity, there are limitations. ROUGE and BLEU might not accurately reflect the quality of abstractive summaries that rephrase the text because of exact word matching (Zhao et al., 2023). More advanced metrics like BERTScore (Zhang et al., 2020) have been proposed to overtake these limitations. BERTScore uses transformer-based models to enumerate a similarity score between the reference and the generated summary based on their contextual embeddings, providing a more refined evaluation that accounts for semantic meaning. MoverScore is another metric that evaluates text similarity by considering both word-level and sentence-level embeddings (Zhao et al., 2023). These newer metrics help provide a more comprehensive assessment of a summary's quality, beyond simple word overlap. We also have a human evaluation which is best because humans can detect hallucinations in text which ATS metrics overlook (Fabbri et al., 2021), but that is time-consuming, so it is required to have continuous improvement in automatic evaluation metrics.

2.5. Future Directions and Trends

One of the coming out trends is the integration of multi-modal summarization which involves summarizing information from multiple sources, such as combining textual data with images or videos to create a coherent summary (Li et al., 2024). This approach can be particularly useful in scenarios like news summarization, where visual content complements textual information. The next growing area of interest is personalized summarization, where the system tailors the generated summary based on the user's preferences, background knowledge, or specific requirements (Wang H et al., 2023). This involves building more interactive summarization models that can take user feedback into account in real-time. While still in its infancy, personalized summarization has the potential to significantly improve user satisfaction and summary relevance.

3. Research Methodology

The methodology for this study is designed to explore Automatic Text Summarization (ATS) using a Modular Pipeline Approach integrated with LangChain, a framework that facilitates easy model management and integration. The methodology will include the following steps: research design, data collection, model selection, implementation process, ethical considerations and evaluation techniques (discussed in the literature review). Each step will ensure that the system is optimized for performance and reliability.

3.1. Research Design

The research embraces a quantitative and experimental design, focusing on the development, implementation, and evaluation of an extensible ATS system. The experimental design involves creating a pipeline that integrates pre-trained models for both extractive and abstractive summarization. This pipeline will be evaluated on its performance across multiple datasets, comparing the results of the modular approach with standalone extractive and abstractive models. LangChain will serve as the framework for managing the modular pipeline and for integrating different models and techniques at various stages of the summarization process (Li et al., 2023).

The study will be conducted in several phases:

- Data collection.
- Model implementation and pipeline development.
- Performance evaluation and analysis.

3.2. Data Collection

The research will use benchmark datasets widely accepted in ATS research to ensure that results are comparable with existing models. These datasets include:

- CNN/DailyMail (Hermann et al., 2015), which consists of news articles and associated summaries.

- XSum (Narayan et al., 2018), another news summarization dataset known for its abstractive properties.
- Scientific Papers Dataset (Cohan et al., 2018), which includes scientific articles, allowing us to assess the models' performance in a technical and domain-specific setting.

The use of these varied datasets allows the modular pipeline to be tested across different domains (news and scientific articles) and different summarization tasks (extractive, abstractive, and hybrid). All datasets will be pre-processed, tokenized, and divided into training, validation, and test sets for accurate evaluation.

3.3. Model Selection

Choosing the right model that has demonstrated efficacy in producing clear and succinct summaries is a crucial decision. Transformer-based architectures make it easy to manage large-scale text data because of their attention mechanisms, enabling them to concentrate on various sections of input text. BART and T5 are excellent options for this task.

3.4. Implementation of the Modular Pipeline

The primary tool used for building the modular summarization pipeline is LangChain, an open-source framework designed to facilitate the chaining of different language models and tasks. The pipeline will combine:
Extractive summarization models to first select the most important sentences or phrases from the text. Abstractive summarization models (like BART or T5) to rephrase and refine these extracted elements into coherent, human-like summaries (Lewis et al., 2020; Raffel et al., 2020). LangChain allows for flexibility in combining multiple models, meaning that the pipeline can be optimized to switch between extractive and abstractive techniques based on the complexity of the input text and the required output quality (Zhang et al., 2024). Moreover, Hugging Face's pre-trained models will be used to accelerate the prototyping process. This library offers a vast range of models trained on specific tasks like summarization, which reduces the need for extensive model training from scratch (Li et al., 2023). By leveraging these models, the system can rapidly test and evaluate different approaches.

3.5. Ethical Considerations

No human subjects are involved in the primary stages of the research. However, human evaluators will be employed to manually assess summaries. All human evaluations will be conducted with informed consent, and participants will remain anonymous. The datasets used are publicly available and do not include sensitive or private information.

4. Research Timeline

The Gantt chart in Figure 5 visually represents the timeline and key milestones for a project focusing on Automatic Text Summarization Using a Modular Pipeline Approach with LangChain. Spanning from September 2024 to January 2025, the chart details the tasks involved in the research and development process. Each task is assigned a specific duration, showcasing how different phases overlap to ensure efficient project management.

The literature review, the longest task at 98 days, runs parallel to other activities such as proposal writing and pipeline design. This reflects the ongoing need to gather relevant research while other aspects of the project progress. Proposal submission is scheduled early on to ensure that the project gains approval before significant technical work begins. Critical tasks like model selection, training, and LangChain integration occur sequentially, ensuring that each component of the system is properly set up and tested before moving forward.

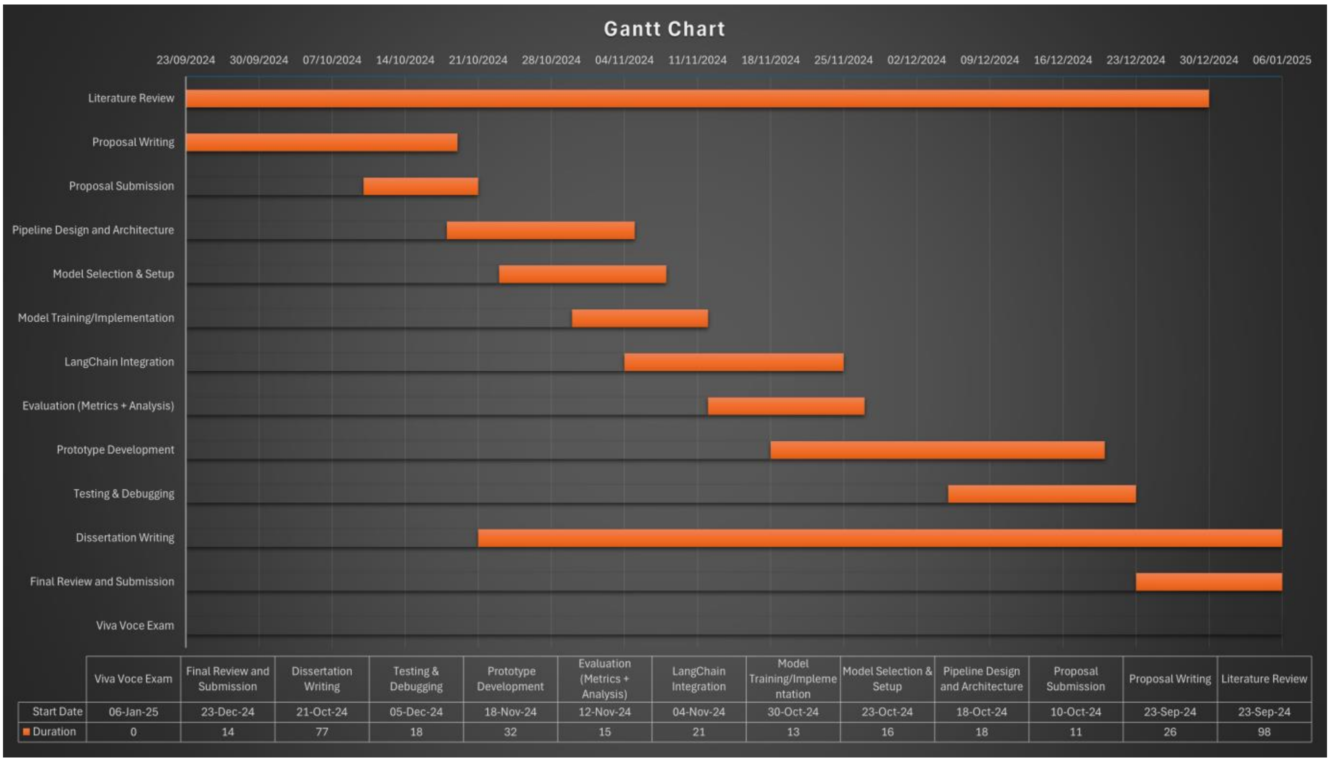

Figure 5 Gantt Chart for Research Timeline

The project emphasizes evaluation and debugging towards the latter stages to ensure the system functions as intended, with metrics analysis running in parallel to prototype development. Finally, the dissertation writing is integrated into the timeline early, allowing for continuous documentation of findings and methodology as well as preparing the viva voce exam.

5. Expected Outcomes

The project on Automatic Text Summarization Using a Modular Pipeline with LangChain aims to enhance summarization quality through BART and T5 models, focusing on both extractive and abstractive methods. The modular pipeline structure allows flexible experimentation with various components, potentially leading to scalable and adaptable summarization solutions. Evaluation metrics like ROUGE and BLEU will quantitatively assess the performance, while human evaluation will gauge readability and coherence. If successful, the system could support real-world applications in media, legal, and customer service, with a prototype serving as a foundation for further research on modular pipelines.

6. Limitations

While the modular pipeline for automatic text summarization using LangChain offers flexibility and scalability, it faces several limitations. The performance is heavily dependent on the dataset, and generalization to other domains, such as legal or scientific texts, is challenging. Managing complex pipeline components and ensuring real-time application suitability are significant challenges, as is extending the system to non-English languages as well as high computational cost can limit experimentation.

7. Conclusion

In conclusion, this research explores the potential of automatic text summarization using a modular pipeline approach with LangChain, leveraging state-of-the-art models like BART and T5. The flexibility of the modular design allows for easy experimentation with different components, enabling adaptability across various text types and summarization methods. While the project is expected to yield improvements in summarization accuracy and scalability, certain challenges—such as the risk of hallucination in abstractive models, computational demands, and limitations of evaluation metrics—remain. Human evaluation will play a crucial role in balancing the quantitative results provided by automated metrics. The modular pipeline also offers scalability and opportunities for future extensions, such as multilingual support and real-time applications. Ultimately, this research aims to enhance the efficacy of summarization systems, contributing to real-world applications in fields like journalism, legal analysis, and customer service while acknowledging its limitations for future improvement.

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